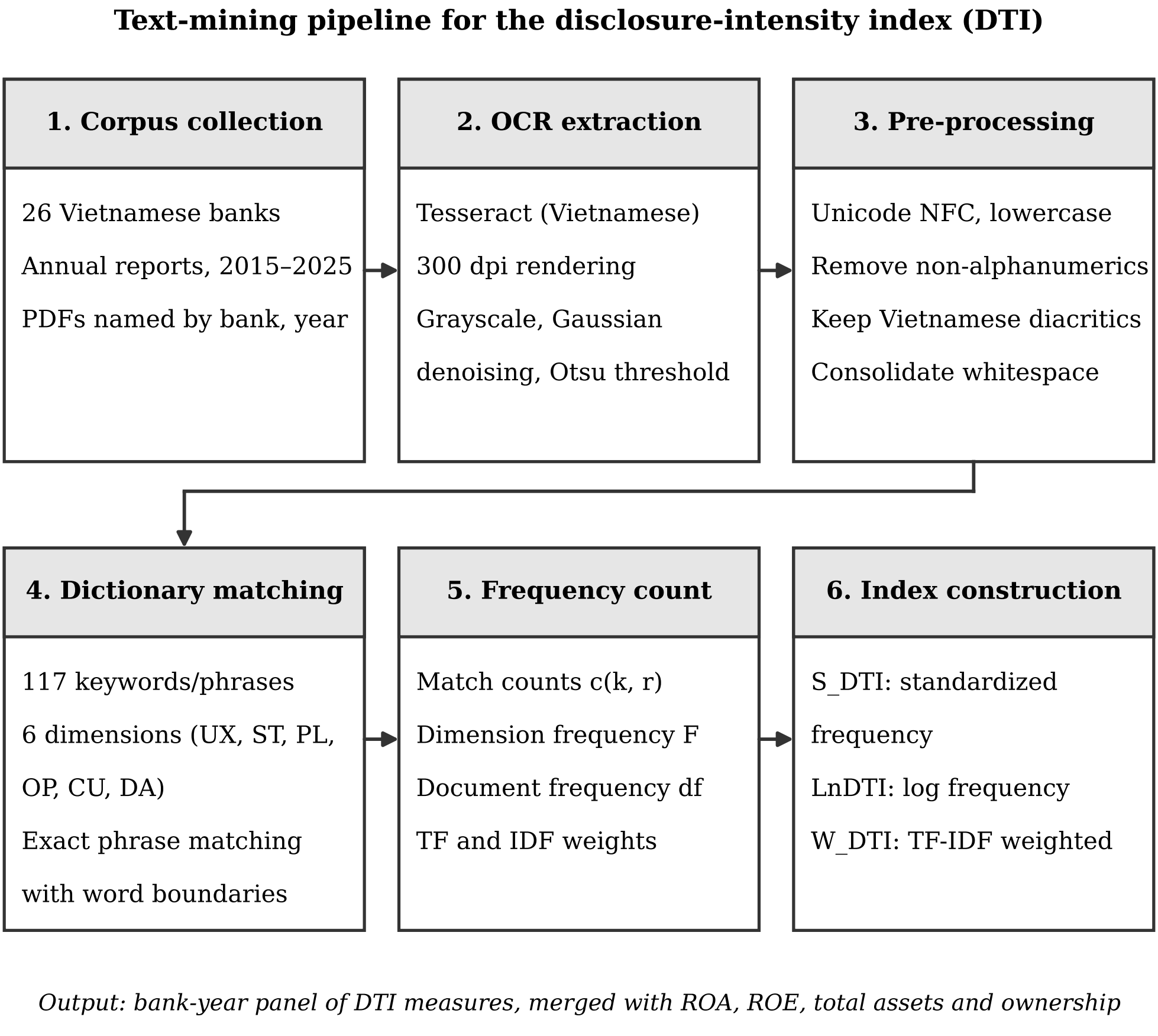
Text-mining pipeline for the disclosure-intensity index (DTI)
1. Corpus collection
26 Vietnamese banks
Annual reports, 2015–2025
PDFs named by bank, year
2. OCR extraction
Tesseract (Vietnamese)
300 dpi rendering
Grayscale, Gaussian
denoising, Otsu threshold
3. Pre-processing
Unicode NFC, lowercase
Remove non-alphanumerics
Keep Vietnamese diacritics
Consolidate whitespace
4. Dictionary matching
117 keywords/phrases
6 dimensions (UX, ST, PL,
OP, CU, DA)
Exact phrase matching
with word boundaries
5. Frequency count
Match counts c(k, r)
Dimension frequency F
Document frequency df
TF and IDF weights
6. Index construction
S_DTI: standardized
frequency
LnDTI: log frequency
W_DTI: TF-IDF weighted
Output: bank-year panel of DTI measures, merged with ROA, ROE, total assets and ownership